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Advanced Synthesis & Catalysis

Synthesis of Tricyclic Isoquinoline Derivatives *via* Palladium-Catalyzed Tandem Reactions of 2,7-Alkadiynylic Carbonates with 2,3-Allenyl Sulfamides

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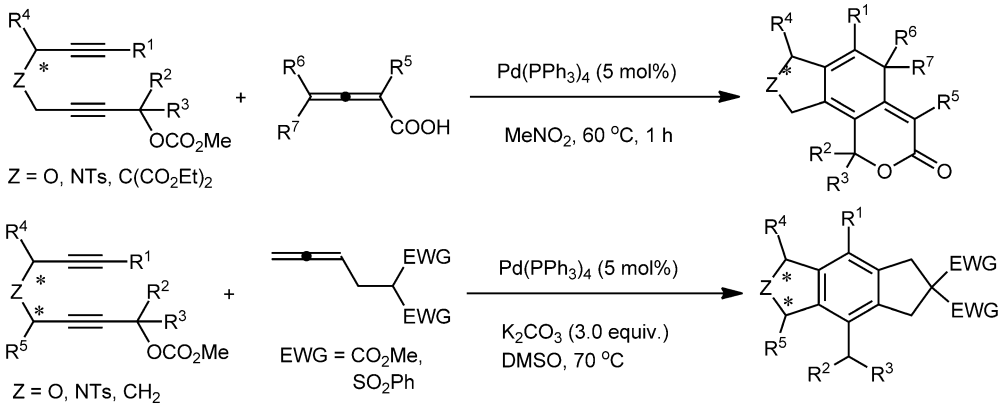
Supporting information for this article is available on the WWW under http://dx.doi.org/10.1002/adsc.201600406.

Abstract: The palladium-catalyzed tandem reactions of 2,7-alkadiynylic carbonates with 2,3-allenyl sulfamides for the syntheses of fused tricycles such as 1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline, 2,3,6,7,8,9-hexahydro-1*H*-pyrrolo[3,4-*h*]isoquinoline, and 2,3,4,7,8,9-hexahydro-1*H*-cyclopenta[*h*]isoquinoline derivatives have been developed. A mechanism has been proposed based on the isolation of a by-product.

Keywords: 2,7-alkadiynylic carbonates; 2,3-allenyl sulfamides; palladium; tandem reactions; tricyclic isoquinoline derivatives

Isoquinoline derivatives are important structural motifs that occur in biologically active alkaloids with antitumor, antimalarial, or anti-inflammatory activities.[1] They are also found in a wide variety of synthetically and functionally valuable compounds such as phosphorescent materials,[2] chiral ligands,[3] and fluorosensors.[4] Therefore, many efforts have been made to develop efficient methods for the construction of isoquinoline skeletons.[5] Nowadays transition metal-catalyzed cyclization reactions of functionalized allenes have become very useful methodologies in organic synthesis.[6] In our previous work, the palladium-catalyzed tandem reactions of 2,7-alkadiynylic carbonates with 2,3-allenoic acids[7] or allenes bearing a carbon-center nucleophile[8] to form fused tricyclic skeletons have been reported (Scheme 1).

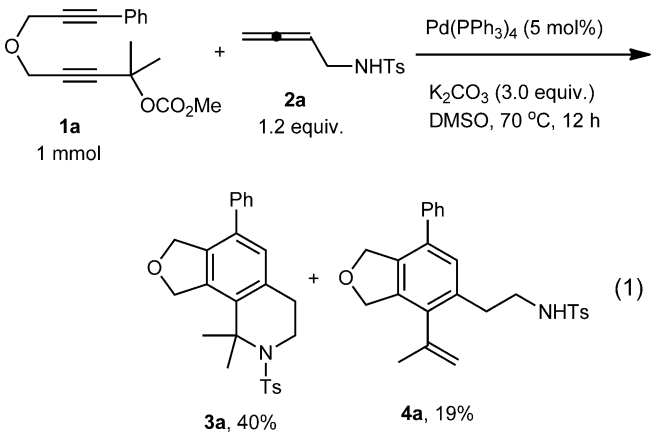
We envisioned that the tricyclic nitrogen-containing heterocycles could be constructed if the 2,3-allenyl sulfamides were applied instead. With this notion in mind, we initially conducted the reaction of 2,7-alkadiynylic carbonate **1a** (1.0 mmol) and *N*-(buta-2,3-dien-1-yl)-4-methylbenzenesulfonamide **2a** (1.2 equiv.) under the catalysis of $Pd(PPh_3)_4$ (5 mol%) and K_2CO_3 (3.0 equiv.) at 70 °C for 12 h in DMSO.[8] To our delight, we successfully isolated a tricyclic 1,3,6,7,8,9-



Scheme 1. The tandem reactions of 2,7-alkadiynylic carbonates with 2,3-allenoic acids or allenes bearing a carbon-centered nucleophile.

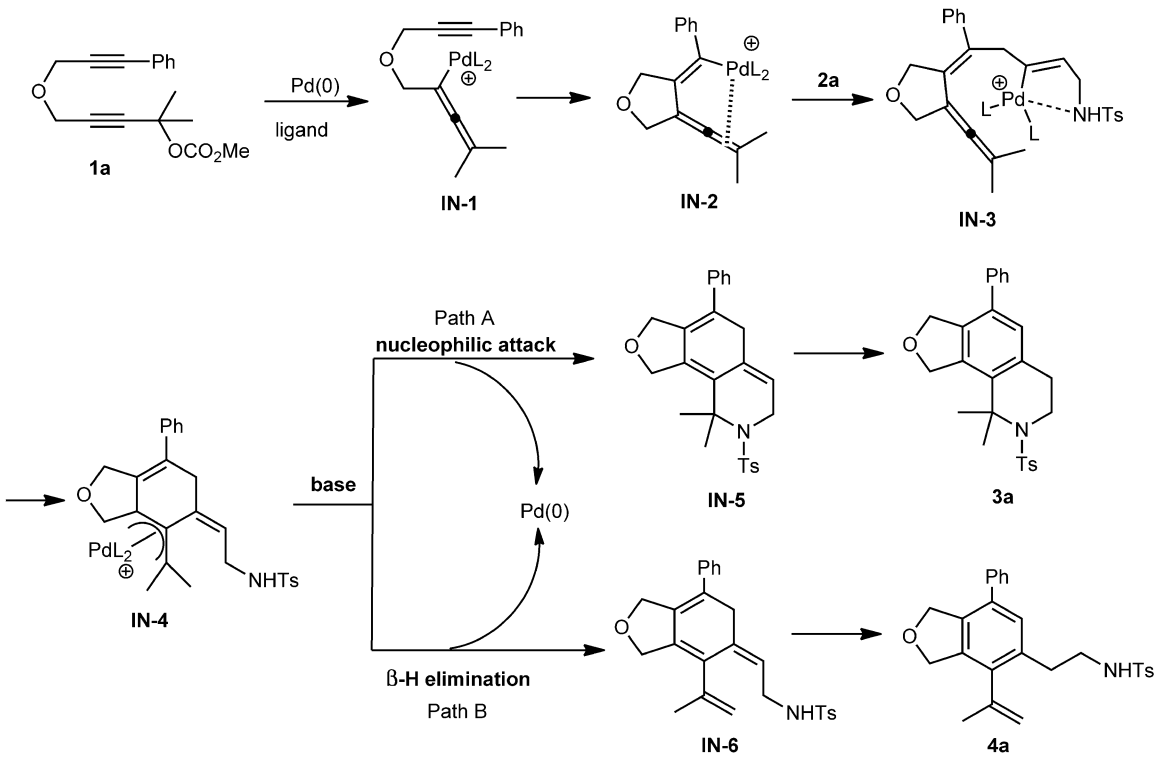
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hexahydrofuro[3,4-*h*]isoquinoline derivative **3a** in 40% yield, together with a benzofuran isomer **4a**, in which the sulfamide unit was kept untouched, in 19% yield [Eq. (1)].



Based on the discovery of **4a**, we propose a possible mechanism for the formation of **3a** and **4a** as shown in Scheme 2. Oxidative addition of **1a** with Pd(0) would afford the allenylpalladium intermediate **IN-1**,[9] which undergoes intramolecular insertion of the triple bond to generate the alkenylpalladium species **IN-2**.[10] The species **IN-2** undergoes highly regioselective intermolecular carbopalladation of the terminal C=C bond of the allene moiety in **2a** to generate a new stereodefined alkenylpalladium intermediate **IN-3** due to the coordination of nitrogen with palladium. Then intramolecular carbopalladation of allene would form a new allylpalladium intermediate **IN-4**. In the presence of a base, the key intermediate **IN-4** could undergo intramolecular nucleophilic attack forming tricyclic intermediate **IN-5** or the β-H elimination forming bicyclic intermediate **IN-6** and regenerating the catalytically active Pd(0). After aromatization, the final products **3a** and **4a** were delivered. The isolation of product **4a** supports this mechanism, which is different from what has been reported in ref.[7]

We then turned to optimize the reaction conditions to improve the yield of **3a** and inhibit the formation of **4a** (Table 1). The reaction using tris(2-furyl)phosphine (TFP) as ligand could afford **3a** in 60% yield with a ratio of **3a**/**4a** = 83/17 (entry 1). No product **3a** was observed with tricyclohexylphosphine (PCy_3), but the β-H elimination product **4a** was observed in 14% yield (entry 2). Further solvent screening showed that the reaction in DMF, DMA, or CH_3NO_2 delivered poorer results than in DMSO (entries 3–5). Interestingly, when CH_3CN was used as the solvent, the tricyclic product **3a** was obtained in 38% yield, however with a much higher ratio of **3a**/**4a** (96/4) (entry 6). It was surprising to find that the reaction could afford **3a** in 84% yield with an excellent ratio of **3a**/**4a** (97/3)

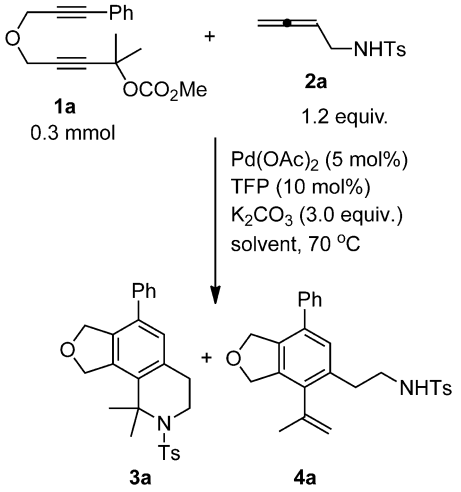


Scheme 2. Proposed mechanism for the formation of **3a** and **4a**.

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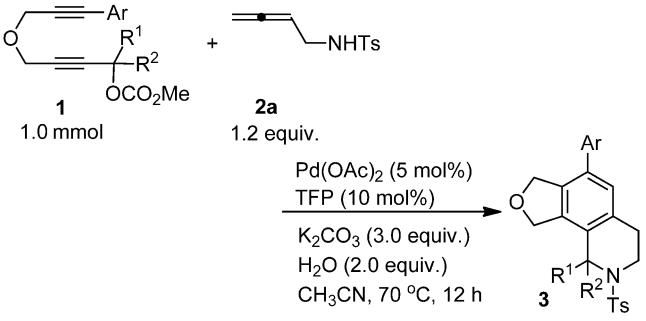
Table 1. Optimization of the reaction conditions.[a]



Entry	Solvent	Time [h]	Yield of **3a** [%][b]	**3a**/**4a**[b]
1	DMSO	12	60	83/17
2[c]	DMSO	12	0	0/100
3	DMF	16	48	67/33
4	DMA	16	39	57/43
5	CH_3NO_2	16	24	81/19
6	CH_3CN	24	38	96/4
7[d]	**CH_3CN**	**11**	**84**	**97/3**
8[e]	CH_3CN	11	82	97/3

[a] *Reaction conditions:* **1a** (0.3 mmol), **2a** (1.2 equiv.), $Pd(OAc)_2$ (5 mol%), TFP (10 mol%), and K_2CO_3 (3.0 equiv.) at 70 °C in solvent (3.0 mL) unless otherwise noted.
[b] Determined by ^1H NMR analysis of the crude products with mesitylene as the internal standard.
[c] PCy_3 was used as the ligand and **4a** was observed in 14% yield.
[d] 2.0 equiv, of H_2O were added.
[e] 6.0 equiv, of H_2O were added.

Table 2. The scope of 2,7-alkadiynylic carbonates **1**.[a]



Entry	**1** Ar	R^1	R^2	Yield of **3** [%][b]
1	Ph	Me	Me (**1a**)	79 (**3a**)
2[c]	Ph	Me	Me (**1a**)	83 (**3a**)
3[d]	Ph	-$(CH_2)_4$- (**1b**)		34 (**3b**)
4	Ph	Me	Pr (**1c**)	61 (**3c**)
5	Ph	Pr	H (**1d**)	50 (**3d**)
6	Ph	Ph	H (**1e**)	52 (**3e**)
7[e]	*p*-ClC_6H_4	Me	Me (**1f**)	71 (**3f**)
8	*p*-$MeOC_6H_4$	Me	Me (**1g**)	66 (**3g**)
9[e]	*p*-MeC_6H_4	Me	Me (**1h**)	66 (**3h**)
10	*o*-MeC_6H_4	Me	Me (**1i**)	60 (**3i**)
11	*m*-MeC_6H_4	Me	Et (**1j**)	60 (**3j**)
12	2-thienyl	Pr	H (**1k**)	45 (**3k**)

[a] *Reaction conditions:* **1** (1.0 mmol), **2a** (1.2 equiv.) $Pd(OAc)_2$ (5 mol%), TFP (10 mol%), K_2CO_3 (3.0 equiv.) and H_2O (2.0 equiv.) at 70 °C in CH_3CN (10.0 mL) for 12 h unless otherwise noted.
[b] Isolated yield.
[c] The reaction was conducted on a one gram scale and the β-H elimination product **4a** was isolated in 3% yield.
[d] The β-H elimination product **4b** was isolated in 28% yield.
[e] The reaction time was 16 h.

when 2.0 equiv. of water were added (entry 7). Increasing the loading of water led to a slightly poorer result (entry 8). We reasoned that the solubility of K_2CO_3 in water increases the efficiency of deprotonation to generate the sulfamide anion, since it was noted that the selectivity of **3a**/**4a** is almost the same in the absence of water: the difference lies in the yield (compare entry 6 with entry 7).[11] Thus, we defined the reaction conditions in entry 7 of Table 1 as optimal for further study.

We first examined the reactivity of various oxygen-tethered 2,7-alkadiynylic carbonates **1** with **2a** affording different 1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline derivatives (Table 2). The reaction of tertiary or secondary 2,7-alkadiynylic carbonates (**1a**–**1e**) proceeded smoothly affording the corresponding tricycles **3a**–**3e** in the yields of 34–83% (entries 1–6). The substrates with the Ar group at the 8-position bearing either electron-deficient or electron-rich groups (**1f**–**1j**) could also be applied (entries 7–11). In addition, this method could be extended to 2-thienyl-substituted 2,7-alkadiynylic carbonate **1k** (entry 12). It is worthy of mention that the reaction could be easily conducted on a one gram scale (entry 2).

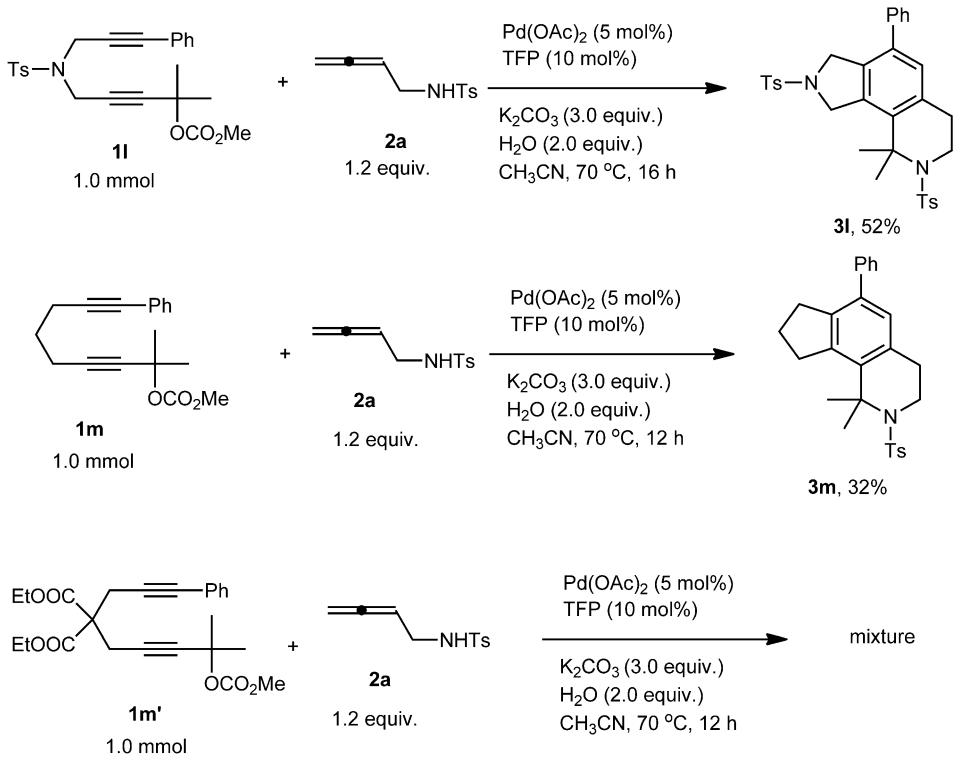
It is also applicable to nitrogen- or carbon-tethered 2,7-alkadiynylic carbonates to form 2,3,6,7,8,9-hexahydro-1*H*-pyrrolo-[3,4-*h*]isoquinoline derivative **3l** in 52% yield and 2,3,4,7,8,9-hexahydro-1*H*-cyclopenta[*h*]isoquinoline derivative **3m** in 32% yield, respectively (Scheme 3). However, when **1m'** was used as the substrate, a complicated mixture was obtained and no defined product could be isolated.

The structure of this tricyclic product was further confirmed on the basis of an X-ray single-crystal diffraction study of the product **3e** (Figure 1).[12]

When the optically active 2,7-alkadiynylic carbonates (*R*)-**1n** and (*R*)-**1o** having a central chirality were applied under the standard reaction conditions, the optically active products (*R*)-**3n** and (*R*)-**3o** were

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Scheme 3. Synthesis of **3l** and **3m**.

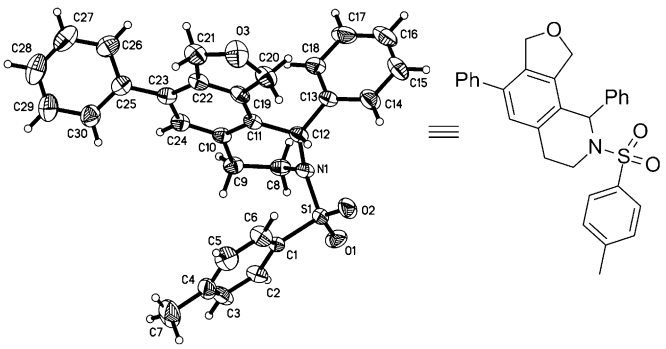


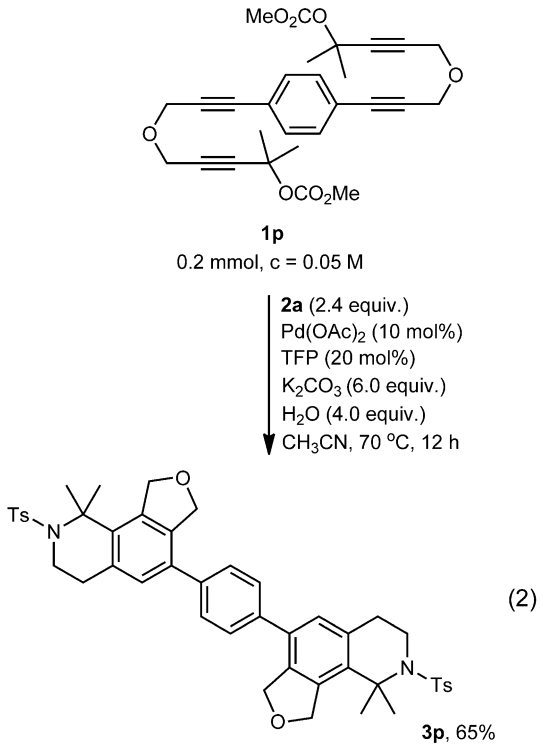
Figure 1. ORTEP representation of product **3e**.

obtained in 76% and 49% yields, respectively, without any loss of enantiomeric purity (Scheme 4).

Interestingly, the substrate **1p**, which has two symmetrical yne-propargylic carbonate units, could also react with **2a** affording the expecting product **3p** in 65% yield with six rings constructing in just one operation [Eq. (2)]!

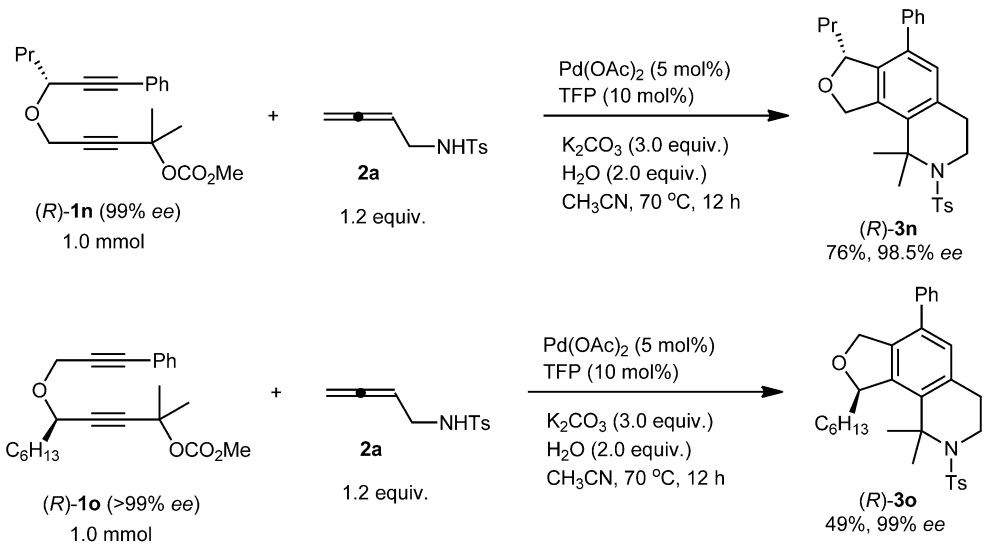
We next explored the scope of 2,3-allenyl sulfamides under the optimized condition (Table 3). The 2,3-allenyl sulfamides bearing a phenyl group at the 1-position or 2-position could react with **1a** affording the tricyclic isoquinoline derivatives **3q** and **3r** in moderate yields (entries 1 and 2). The R^7 group could be either an aryl or alkyl group (entries 3–5).

In conclusion, we have developed an efficient methodology for the syntheses of fused tricycles such as 1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline,



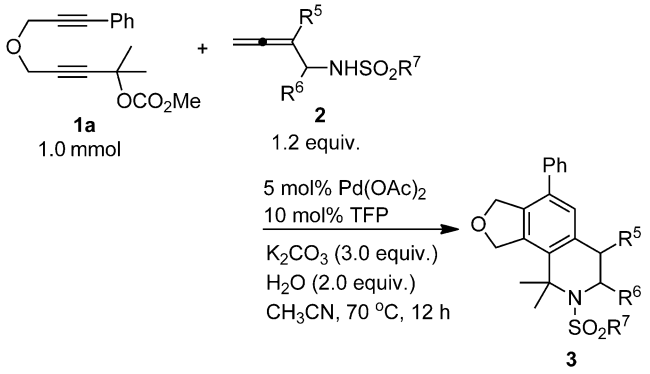
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Scheme 4. Synthesis of optically active tricyclic products (*R*)-**3n** and (*R*)-**3o**.

Table 3. The scope of 2,3-allenyl sulfamides **2**.[a]



Entry	R[5]	R[6]	**2** R[7]	Yield of **3** [%][b]
1	H	Ph	*p*-MeC$_6$H$_4$ (**2b**)	51 (**3q**)
2[c]	Ph	H	*p*-MeC$_6$H$_4$ (**2c**)	52 (**3r**)
3	H	H	Ph (**2d**)	72 (**3s**)
4	H	H	*o*-MeC$_6$H$_4$ (**2e**)	59 (**3t**)
5	H	H	Me (**2f**)	55 (**3u**)

[a] *Reaction conditions:* **1a** (1.0 mmol), **2** (1.2 equiv.) $Pd(OAc)_2$ (5 mol%), TFP (10 mol%), K_2CO_3 (3.0 equiv.) and H_2O (2.0 equiv.) at 70 °C in CH_3CN (10.0 mL) for 12 h unless otherwise noted.
[b] Isolated yield.
[c] The reaction time was 15 h.

2,3,6,7,8,9-hexahydro-1*H*-pyrrolo-[3,4-*h*]isoquinoli-ne, and 2,3,4,7,8,9-hexahydro-1*H*-cyclopenta[*h*]isoquinoline derivatives from 2,7-alkadiynylic carbonates with 2,3-allenyl sulfamides. Three new C–C bonds and a C–N bond are formed in just one operation. The central chirality in the starting materials could be totally transferred into the product. Further studies on synthetic applications are being pursued in our laboratory.

Experimental Section

General Experimental Methods

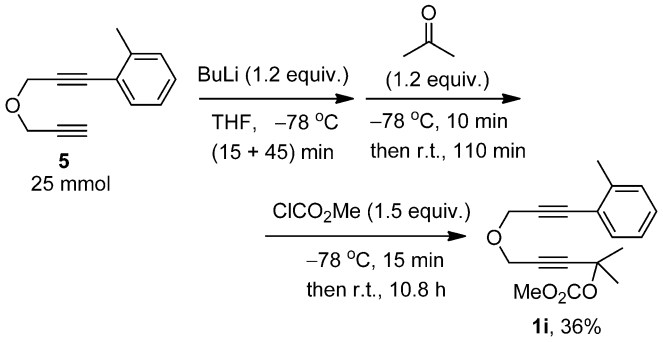
^{1}H and ^{13}C nuclear magnetic resonance spectra were recorded on an instrument operated at 300 MHz for ^{1}H NMR and 75 MHz for ^{13}C NMR spectra. Infrared spectra were recorded from the films of pure samples on sodium chloride plates for liquid or in the form of KBr discs for the solid samples. Mass and HR-mass spectra were carried out in EI or ESI mode. Thin layer chromatography was performed on precoated glass-backed plates and visualized with UV light at 254 nm. K_2CO_3 was bought from Sinopharm Chemical Reagent Co., Ltd. $Pd(OAc)_2$ was bought from Acros. TFP was bought from *J&K*. CH_3CN used was dried over CaH_2 at reflux conditions and distilled just before use. All the temperatures are referred to the oil baths used. The synthesis of starting material 2,7-alkadiynylic carbonates **1** was reported in our previous work.[8]

Synthesis of Methyl {2-methyl-5-[(3-{*o*-tolyl]prop-2-yn-1-yl)oxy]pent-3-yn-2-yl] Carbonate (1i)

To a solution of **5** (4.5985 g, 25 mmol) in THF (80 mL) was added dropwise *n*-BuLi (2.5 M in hexane, 12.0 mL, 30 mmol) at −78 °C within 15 min. After lithiation for 45 min at −78 °C, acetone (2.2 mL, d = 0.80 g mL^{-1}, 1.76 g, 30.3 mmol) was added dropwise with an addition funnel at this temperature within 10 min. The cooling bath was removed and the reaction mixture was warmed up to room temperature and stirred for 110 min. Then methyl chloroformate (2.9 mL, d = 1.22 g mL^{-1}, 3.54 g, 37.5 mmol) was added dropwise at −78 °C in 15 min. After that the cooling bath was removed and the reaction mixture was allowed to warm

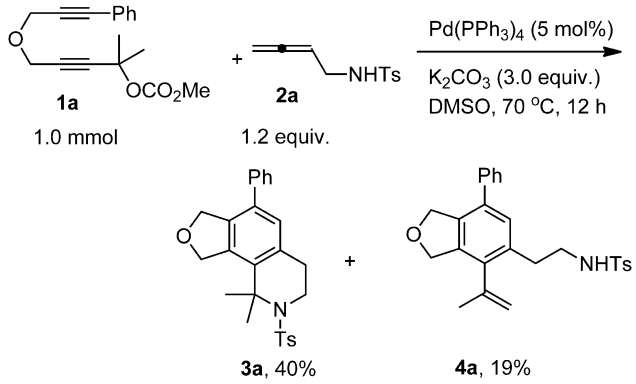
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up to room temperature gradually with stirring for 10.8 h. After the reaction was complete as monitored by TLC [eluent: petroleum ether/ethyl acetate (20/1)], it was quenched with an aqueous solution of saturated NH_4Cl (30 mL). The mixture was extracted with ethyl acetate (30 mL × 2), and the combined organic phase was dried over anhydrous Na_2SO_4. After filtration, evaporation of the solvent and chromatography on silica gel [eluent: petroleum ether/ethyl acetate (50/1)] afforded **1i** as a liquid; yield: 2.7219 g (36%). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.41 (d, J = 7.2 Hz, 1H, ArH), 7.26–7.08 (m, 3H, ArH), 4.52 (s, 2H, CH_2), 4.38 (s, 2H, CH_2), 3.76 (s, 3H, Me), 2.43 (s, 3H, Me), 1.72 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 153.4, 140.3, 132.1, 129.3, 128.5, 125.5, 122.2, 88.0, 87.2, 85.6, 79.9, 74.0, 57.2, 56.5, 54.3, 28.7, 20.6; IR (neat): n = 3063, 2990, 2954, 2852, 2233, 2203, 1754, 1486, 1441, 1384, 1365, 1350, 1275, 1194, 1137, 1083 cm^{-1}; MS (EI): m/z (%) = 300 (M^+, 2.0), 195 (100); HR-MS: m/z = 300.1360, calcd. for $C_{18}H_{20}O_4$ (M^+): 300.1362.

Reaction of 1a and 2a using $Pd(PPh_3)_4$ as the Catalyst[8]

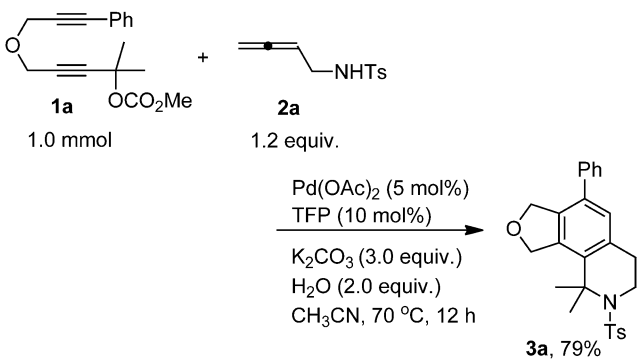


To a flame-dried Schlenk tube containing K_2CO_3 (414.1 mg, 3.0 mmol) were added $Pd(PPh_3)_4$ (57.9 mg, 0.05 mmol), **2a** (268.5 mg, 1.2 mmol), and **1a** (288.0 mg, 1.0 mmol)/DMSO (10.0 mL) sequentially under a nitrogen atmosphere. The reaction was complete after being stirred at 70 °C for 12 h as monitored by TLC (eluent: petroleum ether/ethyl acetate = 5/1). After cooling to room temperature, the resulting mixture was quenched with an aqueous solution of diluted hydrochloric acid [(v/v = 1/10, commercially available concentrated hydrochloric acid was diluted with water according to the volume ratio), 10 mL] slowly and extracted with ethyl acetate (30 mL × 3). The combined organic layer was washed with water and brine, and dried over anhydrous Na_2SO_4. After filtration, evaporation of the solvent, and chromatography on silica gel [eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1)] afforded a part of pure **3a**, a part of pure **4a**, and a mixture of **3a** and **4a** with impurities. The mixture was further purified with chromatography on silica gel [eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1)] to yield another part of pure **3a** and another part of pure **4a**); products **3a** (yield: 175.2 mg, 40%) and **4a** (yield: 84.6 mg, 19%) were obtained, respectively.

3a: solid; mp 188–189 °C (n-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, J = 8.4 Hz, 2H, ArH), 7.46–7.23 (m, 7H, ArH), 7.03 (s, 1H, ArH), 5.23 (s, 2H, CH_2), 5.05 (s, 2H, CH_2), 3.70 (t, J = 5.6 Hz, 2H, CH_2), 2.87 (t, J = 5.4 Hz, 2H, CH_2), 2.40 (s, 3H, Me), 1.85 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 142.9, 140.2, 139.4, 138.0, 137.5, 136.0, 134.0, 133.9, 129.5, 129.1, 128.6, 127.8, 127.5, 126.8, 74.0, 72.4, 62.2, 42.1, 31.0, 27.1, 21.4; IR (KBr): ν = 3064, 3000, 2942, 2907, 2861, 2831, 1596, 1468, 1447, 1397, 1369, 1323, 1303, 1161, 1126, 1089, 1060 cm^{-1}; MS (EI): m/z (%) = 433 (M^+, 0.5), 418 (100); anal. calcd. for $C_{26}H_{27}NO_3S$ (%): C 72.03, H 6.28, N 3.23; found: C 71.67, H 6.30, N 3.02.

4a: liquid; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.72–7.64 (m, 2H, ArH), 7.47–7.30 (m, 5H, ArH), 7.24–7.17 (m, 2H, ArH), 7.03 (s, 1H, ArH), 5.24–5.15 (m, 3H, one proton of =CH_2 and CH_2), 5.05–4.98 (m, 2H, CH_2), 4.88 (t, J = 6.2 Hz, 1H, NH), 4.78–4.73 (m, 1H, one proton of =CH_2), 3.25–3.12 (m, 2H, CH_2), 2.80 (t, J = 7.4 Hz, 2H, CH_2), 2.37 (s, 3H, Me), 1.92 (s, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 143.3, 142.5, 139.6, 138.3, 136.8, 136.5, 135.4, 134.6, 134.3, 129.6, 128.7, 128.6, 127.7, 127.5, 127.0, 116.2, 73.7, 73.1, 44.3, 32.4, 23.6, 21.4; IR (neat): ν = 3284, 3057, 3027, 2932, 2856, 1598, 1475, 1446, 1328, 1160, 1094, 1058 cm^{-1}; MS (EI): m/z (%) = 433 (M^+, 1.83), 278 (100); HR-MS: m/z = 433.1703, calcd. for $C_{26}H_{27}NO_3S$ (M^+): 433.1712.

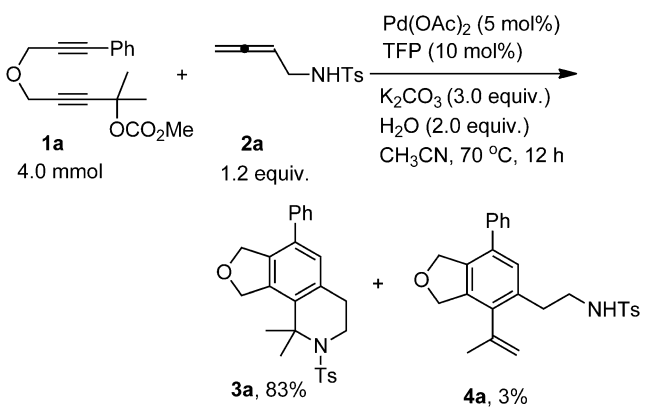
Preparation of 9,9-Dimethyl-4-phenyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3a) and 4-Methyl-*N*-{2-[7-phenyl-4-(prop-1-en-2-yl)-1,3-dihydroisobenzofuran-5-yl]ethyl}benzenesulfonamide (4a)



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Typical Procedure: To a flame-dried Schlenk tube containing K_2CO_3 (413.6 mg, 3.0 mmol) were added $Pd(OAc)_2$ (11.2 mg, 0.05 mmol), TFP (23.0 mg, 0.10 mmol), **2a** (268.2 mg, 1.2 mmol), **1a** (285.1 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.4 mg, 2.0 mmol)/CH_3CN (2.0 mL) sequentially under a nitrogen atmosphere. The reaction was complete after being stirred at 70 °C for 12 h as monitored by TLC (eluent: petroleum ether/ethyl acetate/dichloromethane = 6/1/1). After cooling to the room temperature, the resulting mixture was filtered through a short column of silica gel eluted with ethyl acetate (30 mL). After evaporation of the solvent, the yields of product **3a** and **4a** were determined by ^{1}H NMR analysis of the crude product with mesitylene (46 μL) as the internal standard. Chromatography on silica gel (eluent: petroleum ether/ethyl acetate/dichloromethane = 6/1/1) afforded **3a** as a solid; yield: 340.2 mg (79%). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, J = 8.4 Hz, 2H, ArH), 7.47–7.22 (m, 7H, ArH), 7.03 (s, 1H, ArH), 5.23 (s, 2H, CH_2), 5.06 (s, 2H, CH_2), 3.71 (t, J = 5.4 Hz, 2H, CH_2), 2.87 (t, J = 5.0 Hz, 2H, CH_2), 2.41 (s, 3H, Me), 1.85 (s, 6H, Me × 2).



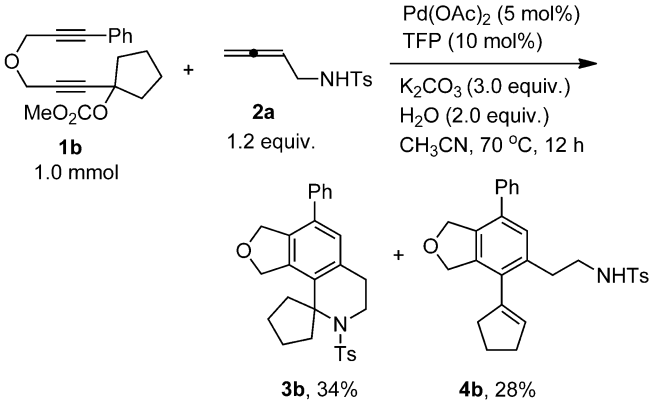
4.0 mmol Scale: To a flame-dried three-neck flask equipped with a reflux condenser containing K_2CO_3 (1.6580 g, 12.0 mmol) were added $Pd(OAc)_2$ (45.0 mg, 0.2 mmol), TFP (92.8 mg, 0.8 mmol), **2a** (1.0728 g, 4.8 mmol), **1a** (1.1487 g, 4.0 mmol)/CH_3CN (30.0 mL), and H_2O (143.8 mg, 8.0 mmol)/CH_3CN (10.0 mL) sequentially under a nitrogen atmosphere. The reaction was complete after being stirred at 70 °C for 12 h as monitored by TLC (eluent: petroleum ether/ethyl acetate = 5/1). After cooling to the room temperature, the resulting mixturewas filtered through a short column of silica gel (ethyl acetate, 80 mL). After evaporation of the solvent, the yields of product **3a** and **4a** were determined by ^{1}H NMR analysis of the crude product with mesitylene (184 μL) as the internal standard. Chromatography on silica gel (eluent: petroleum ether/ethyl acetate/dichloromethane (5/1/1) afforded impure **3a** and **4a**, which were further purified by using the eluent [dichloromethane to dichloromethane/ethyl acetate (50/1) for **3a**] and the eluent [petroleum ether/ethyl acetate/dichloromethane = 6/1/1) for **4a**] affording **3a** (yield: 1.4521 g, 83%) and **4a** (yield: 43.9 mg, 3%).

3a: ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, 2H, J = 8.4 Hz, ArH), 7.46–7.24 (m, 7H, ArH), 7.03 (s, 1H, ArH), 5.23 (s, 2H, CH_2), 5.05 (s, 2H, CH_2), 3.71 (t, J = 5.7 Hz, 2H, CH_2), 2.87 (t, J = 5.4 Hz, 2H, CH_2), 2.41 (s, 3H, Me), 1.86 (s, 6H, Me × 2).

4a: ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.71–7.64 (m, 2H, ArH), 7.48–7.30 (m, 5H, ArH), 7.25–7.18 (m, 2H, ArH), 7.02 (s, 1H, ArH), 5.26–5.16 (m, 3H, one proton of = CH_2 and CH_2), 5.06–4.99 (m, 2H, CH_2), 4.82–4.75 (m, 1H, one proton of = CH_2), 4.49 (brs, 1H, NH), 3.25–3.15 (m, 2H, CH_2), 2.82 (t, J = 7.2 Hz, 2H, CH_2), 2.39 (s, 3H, Me), 1.94 (s, 3H, Me).

The following compounds **3b–3u** were prepared according to the Typical Procedure except for compound **3p**.

Preparation of 4′-phenyl-8′-tosyl-3′,6′,7′,8′-tetrahydro-1′*H*-spiro[cyclopentane-1,9′-furo[3,4-*h*]isoquinoline] (3b) and *N*-{2-[4-(cyclopent-1-en-1-yl)-7-phenyl-1,3-dihydroisobenzofuran-5-yl]ethyl}-4-methylbenzenesulfonamide (4b)



The reaction of K_2CO_3 (414.4 mg, 3.0 mmol), $Pd(OAc)_2$ (11.0 mg, 0.05 mmol), TFP (23.1 mg, 0.10 mmol), **2a** (267.5 mg, 1.2 mmol), **1b** (313.7 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.3 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3b** (yield: 157.4 mg, 3%) as a solid and **4b** (yield: 129.6 mg, 28%) as a solid after chromatography on silica gel [eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1) afforded one part of pure **3b**, one part of pure **4b**, and a mixture of **3b** and **4b**, which was further purified by using chromatography petroleum ether/ethyl acetate/dichloromethane (6/1/1) as the eluent].

3b: mp 202–203 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.57–7.18 (m, 7H, ArH), 7.06 (d, J = 7.8 Hz, 2H, ArH), 6.80 (s, 1H, ArH), 5.18 (s, 2H, CH_2), 5.01 (s, 2H, CH_2), 4.02 (t, J = 6.8 Hz, 2H, CH_2), 2.93 (t, J = 6.6 Hz, 2H, CH_2), 2.81–2.56 (m, 2H, CH_2), 2.26 (s, 3H, Me), 2.05–1.62 (m, 6H, CH_2 × 3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 142.5, 139.5, 138.5, 135.9, 135.3, 134.4, 134.0, 129.2, 128.9, 128.5, 127.6, 127.4, 126.9, 73.5, 72.2, 71.5, 42.3, 38.5, 27.4, 23.4, 21.3; IR (KBr): ν = 2985, 2944, 2918, 2869, 1602, 1575, 1495, 1474, 1447, 1397, 1335, 1307, 1233, 1166, 1155, 1091, 1072 cm^{-1}; MS (EI): m/z (%) = 459 (M^+, 1.90), 304 (100); HR-MS: m/z = 459.1869, calcd. for $C_{28}H_{29}NO_3S$ (M^+): 459.1868.

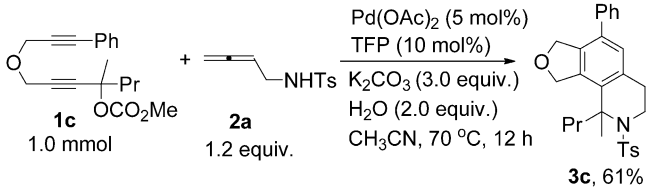
4b: mp 148–150 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.67 (d, J = 7.8 Hz, 2H, ArH), 7.47–7.25 (m, 5H,

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16154169, 2016, 17, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201600406 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

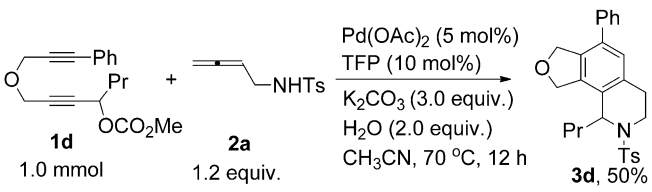
ArH), 7.20 (d, J=8.1 Hz, 2H, ArH), 7.03 (s, 1H, ArH), 5.54 (s, 1H, CH=), 5.16 (s, 2H, CH_2), 5.08 (t, J=5.9 Hz, 1H, NH), 4.97 (s, 2H, CH_2), 3.21–3.06 (m, 2H, CH_2), 2.80 (t, J=7.2 Hz, 2H, CH_2), 2.50–2.30 (m, 7H, CH_2×2 and Me), 2.04–1.88 (m, 2H, CH_2); ^{13}C NMR (75 MHz, $CDCl_3$): δ=143.2, 140.4, 139.5, 139.1, 136.7, 135.4, 135.2, 134.4, 132.1, 130.5, 129.5, 128.7, 128.6, 127.6, 127.4, 126.9, 73.6, 73.3, 44.1, 36.5, 33.1, 32.9, 23.8, 21.3; IR (KBr): ν=3278, 3060, 3027, 2947, 2847, 1599, 1492, 1472, 1446, 1327, 1160, 1094, 1081, 1061, 1040 cm^{-1}; MS (EI): m/z (%)=459 (M^+, 2.18), 304 (100); HR-MS: m/z=459.1872, calcd. for $C_{28}H_{29}NO_3S$ (M^+): 459.1868.

Preparation of 9-Methyl-4-phenyl-9-propyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3c)



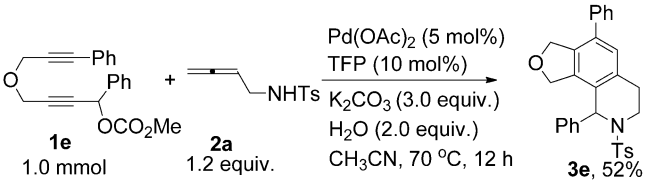
The reaction of K_2CO_3 (414.1 mg, 3.0 mmol), $Pd(OAc)_2$ (11.1 mg, 0.05 mmol), TFP (23.2 mg, 0.10 mmol), **2a** (267.4 mg, 1.2 mmol), **1c** (312.5 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.2 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3c** (yield: 278.0 mg, 61%) as a liquid {eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1) afforded impure **3c**, which was further purified with chromatography [eluent: dichloromethane (200 mL) to dichloromethane/ethyl acetate (50/1)]}. ^{1}H NMR (300 MHz, $CDCl_3$): δ=7.80 (d, J=8.1 Hz, 2H, ArH), 7.47–7.20 (m, 7H, ArH), 7.00 (s, 1H, ArH), 5.28 (d, J=11.7 Hz, 1H, one proton of CH_2), 5.18–4.99 (m, 3H, one proton of CH_2 and CH_2), 3.70–3.45 (m, 2H, CH_2), 2.90–2.70 (m, 3H, one proton of CH_2 and CH_2), 2.39 (s, 3H, Me), 1.92–1.65 (m, 4H, one proton of CH_2 and Me), 1.50–1.24 (m, 1H, one proton of CH_2), 1.00–0.63 (m, 4H, one proton of CH_2 and Me); ^{13}C NMR (75 MHz, $CDCl_3$): δ=142.8, 139.3, 139.2, 137.2, 136.4, 135.4, 135.0, 133.6, 129.3, 128.7, 128.4, 127.6, 127.3, 126.9, 74.1, 72.3, 66.0, 43.6, 41.8, 30.6, 27.4, 21.2, 18.0, 13.9; IR (neat): ν=3063, 3027, 2959, 2929, 2871, 1736, 1598, 1494, 1466, 1400, 1365, 1322, 1268, 1162, 1147, 1129, 1092, 1071, 1018 cm^{-1}; MS (EI): m/z (%)=461 (M^+, 0.56), 418 (100), 91 (100); anal. calcd. for $C_{28}H_{31}NO_3S$ (%): C 72.85, H 6.77, N 3.03; found: C 72.88, H 6.65, N 2.79.

Preparation of 4-Phenyl-9-propyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3d)



The reaction of K_2CO_3 (414.0 mg, 3.0 mmol), $Pd(OAc)_2$ (11.1 mg, 0.05 mmol), TFP (23.3 mg, 0.10 mmol), **2a** (267.4 mg, 1.2 mmol), **1d** (300.3 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.3 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3d** (yield: 224.6 mg, 50%) as a solid {eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1) afforded impure **3d**, which was further purified with chromatography [eluent: dichloromethane (200 mL) to dichloromethane/ethyl acetate (50/1)]}; mp 110–111 °C (n-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ=7.58 (d, J=8.1 Hz, 2H, ArH), 7.47–7.20 (m, 5H, ArH), 7.08 (d, J=7.8 Hz, 2H, ArH), 6.83 (s, 1H, ArH), 5.30–5.00 (m, 4H, one proton of CH_2+CH+CH_2), 4.84 (d, J=9.3 Hz, 1H, one proton of CH_2), 3.98–3.80 (m, 1H, one proton of CH_2), 3.65–3.44 (m, 1H, one proton of CH_2), 2.70–2.53 (m, 2H, CH_2), 2.27 (s, 3H, Me), 1.88–1.68 (m, 1H, one proton of CH_2), 1.67–1.38 (m, 3H, one proton of CH_2 and CH_2), 0.97 (t, J=6.5 Hz, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): δ=142.9, 139.4, 137.2, 136.1, 135.0, 134.0, 132.2, 129.9, 129.1, 128.4, 128.1, 127.4, 127.2, 126.7, 73.2, 72.1, 54.1, 38.2, 36.7, 25.4, 21.2, 19.2, 13.4; IR (KBr): ν=3063, 3029, 2957, 2871, 1598, 1492, 1476, 1406, 1380, 1337, 1305, 1271, 1163, 1114, 1092, 1060, 1018 cm^{-1}; MS (EI): m/z (%)=447 (M^+, 0.05), 404 (100); anal. calcd. for $C_{27}H_{29}NO_3S$ (%): C 72.45, H 6.53, N 3.13; found: C 72.56, H 6.36, N 2.99.

Preparation of 4,9-Diphenyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3e)

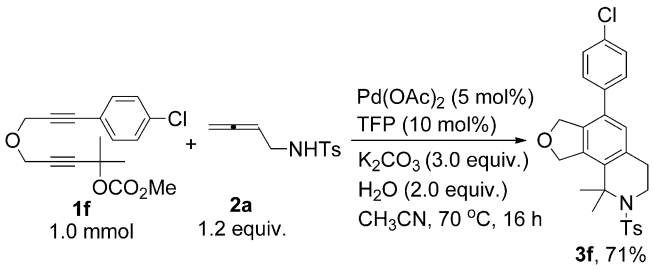


The reaction of K_2CO_3 (414.6 mg, 3.0 mmol), $Pd(OAc)_2$ (11.1 mg, 0.05 mmol), TFP (23.3 mg, 0.10 mmol), **2a** (267.6 mg, 1.2 mmol), **1e** (336.4 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (37.4 mg, 2.1 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3e** (yield: 253.7 mg, 52%) as a solid {eluent: petroleum ether/ethyl acetate/dichloromethane afforded impure **3d**, which was further purified with chromatography [eluent: dichloromethane/ethyl acetate (60/1)]}; mp 169–170 °C (n-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ=7.57 (d, J=8.1 Hz, 2H, ArH), 7.45–7.16 (m, 10H, ArH), 7.09 (d, J=7.8 Hz, 2H, ArH), 7.00 (s, 1H, ArH), 6.10 (s, 1H, CH), 5.20–5.03 (m, 2H, CH_2), 4.96 (d, J=12.9 Hz, 1H, one proton of CH_2), 4.37 (d, J=12.6 Hz, 1H, one proton of CH_2), 3.84 (dd, J_1=14.3 Hz, J_2=6.2 Hz, 1H, one proton of CH_2), 3.38–3.22 (m, 1H, one proton of CH_2), 2.90–2.73 (m, 1H, one proton of CH_2), 2.66 (dd, J_1=16.8 Hz, J_2=3.3 Hz, 1H, one proton of CH_2), 2.28 (s, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): δ=143.0, 139.4, 138.9, 137.6, 137.4, 135.0, 134.6, 133.3, 129.2, 128.5, 128.45, 128.38, 128.3, 127.9, 127.5, 127.4, 126.8, 73.3, 72.3, 57.1, 38.3, 26.3, 21.2; IR (KBr): ν=3060, 3029, 2938, 2848, 1599, 1494, 1477, 1452, 1337, 1267, 1161, 1096, 1060, 1029 cm^{-1}; MS (EI): m/z (%)=481 (M^+, 1.79), 326 (100); anal. calcd. for $C_{30}H_{27}NO_3S$ (%): C 74.82, H 5.65, N 2.91; found: C 74.91, H 5.66, N 2.72.

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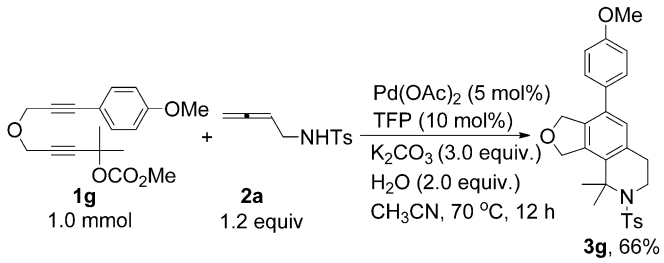
16154169, 2016, 17, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201600406 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Preparation of 4-(4-Chlorophenyl)-9,9-dimethyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3f)



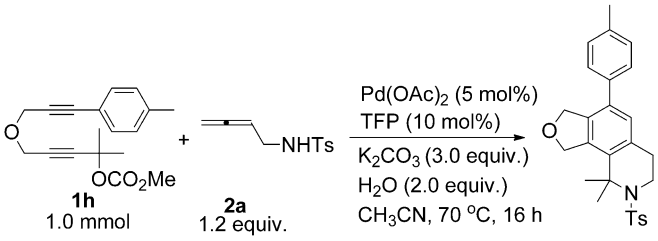
The reaction of K_2CO_3 (414.2 mg, 3.0 mmol), $Pd(OAc)_2$ (11.3 mg, 0.05 mmol), TFP (23.1 mg, 0.10 mmol), **2a** (267.8 mg, 1.2 mmol), **1f** (318.1 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.1 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 16 h afforded **3f** (yield: 330.0 mg, 71%) as a solid after chromatography on silica gel [eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1)]; mp 195–197 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, J = 8.1 Hz, 2H, ArH), 7.37 (d, J = 8.4 Hz, 2H, ArH), 7.32–7.21 (m, 4H, ArH), 7.00 (s, 1H, ArH), 5.22 (s, 2H, CH_2), 5.02 (s, 2H, CH_2), 3.70 (t, J = 5.3 Hz, 2H, CH_2), 2.87 (t, J = 5.0 Hz, 2H, CH_2), 2.40 (s, 3H, Me), 1.84 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 142.9, 140.1, 138.3, 137.7, 137.2, 136.1, 134.0, 133.5, 132.6, 129.4, 129.0, 128.9, 128.7, 126.7, 73.9, 72.1, 62.0, 42.0, 30.9, 27.0, 21.3; IR (KBr): ν = 2942, 2866, 2842, 1597, 1496, 1469, 1429, 1389, 1367, 1320, 1270, 1250, 1151, 1127, 1092, 1065, 1014 cm^{-1}; MS (EI): *m/z* (%) = 469 [M$^+$(^{37}Cl), 0.16], 467 [M$^+$(^{35}Cl), 0.79], 452 (100); anal. calcd. for $C_{26}H_{26}ClNO_3S$ (%): C 66.73, H 5.60, N 2.99; found: C 66.56, H 5.63, N 2.81.

Preparation of 4-(4-Methoxyphenyl)-9,9-dimethyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3g)



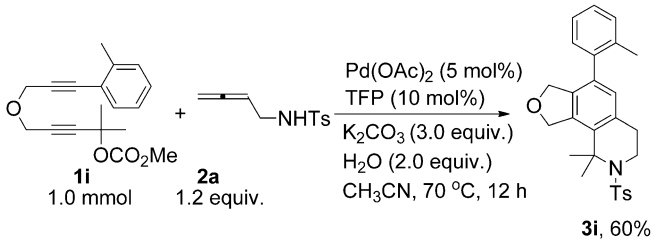
The reaction of K_2CO_3 (415.2 mg, 3.0 mmol), $Pd(OAc)_2$ (11.3 mg, 0.05 mmol), TFP (23.4 mg, 0.10 mmol), **2a** (269.4 mg, 1.2 mmol), **1g** (316.1 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.2 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3g** (yield: 307.7 mg, 66%) as a solid [eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1)]: mp 104–106 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, J = 8.1 Hz, 2H, ArH), 7.27 (d, J = 7.2 Hz, 4H, ArH), 7.00 (s, 1H, ArH), 6.95 (d, J = 8.7 Hz, 2H, ArH), 5.22 (s, 2H, CH_2), 5.05 (s, 2H, CH_2), 3.83 (s, 3H, Me), 3.70 (t, J = 5.4 Hz, 2H, CH_2), 2.87 (t, J = 5.0 Hz, 2H, CH_2), 2.40 (s, 3H, Me), 1.84 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 159.0, 142.9, 140.2, 137.5, 137.3, 135.9, 133.8, 133.6, 131.7, 129.5, 128.9, 126.8, 114.0, 74.0, 72.4, 62.1, 55.2, 42.1, 31.0, 27.1, 21.4; IR (KBr): ν = 3000, 2936, 2837, 1610, 1517, 1469, 1367, 1320, 1250, 1150, 1129, 1110, 1091, 1064, 1032 cm^{-1}; MS (EI): *m/z* (%) = 463 (M$^+$, 4.81), 448 (100); anal. calcd. for $C_{27}H_{29}NO_4S$ (%): C 69.95, H 6.31, N 3.02; found: C 69.84, H 6.21, N 2.81.

Preparation of 9,9-Dimethyl-4-(*p*-tolyl)-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3h)



The reaction of K_2CO_3 (414.5 mg, 3.0 mmol), $Pd(OAc)_2$ (11.2 mg, 0.05 mmol), TFP (23.4 mg, 0.10 mmol), **2a** (266.7 mg, 1.2 mmol), **1h** (297.3 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.3 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 16 h afforded **3h** (yield: 294.0 mg, 66%) as a solid {eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1) afforded impure **3h**, which was further purified with chromatography [eluent: dichloromethane (200 mL) to dichloromethane/ethyl acetate (50/1)]}; mp 152–153 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.77 (d, J = 8.1 Hz, 2H, ArH), 7.30–7.16 (m, 6H, ArH), 7.02 (s, 1H, ArH), 5.21 (s, 2H, CH_2), 5.04 (s, 2H, CH_2), 3.69 (t, J = 5.3 Hz, 2H, CH_2), 2.86 (t, J = 4.5 Hz, 2H, CH_2), 2.37 (s, 3H, Me), 2.36 (s, 3H, Me), 1.84 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 142.7, 140.1, 137.5, 137.2, 137.1, 136.3, 135.8, 133.8, 133.7, 129.3, 129.1, 128.8, 127.4, 126.6, 73.8, 72.2, 61.9, 42.0, 30.8, 26.9, 21.2, 20.9; IR (KBr): ν = 3024, 2980, 2944, 2917, 2861, 1667, 1598, 1518, 1470, 1393, 1365, 1317, 1267, 1250, 1156, 1128, 1091, 1064 cm^{-1}; MS (EI): *m/z* (%) = 447 (M$^+$, 1.20), 432 (100); anal. calcd. for $C_{27}H_{29}NO_3S$ (%): C 72.45, H 6.53, N 3.13; found: C 72.81, H 6.57, N 2.95.

Preparation of 9,9-Dimethyl-4-(*o*-tolyl)-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3i)



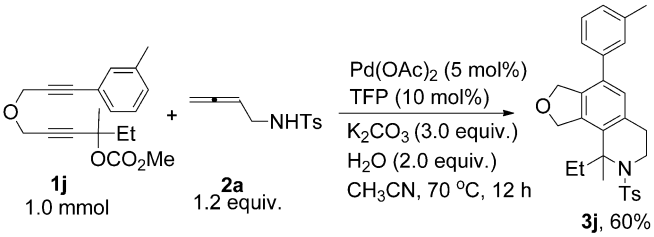
The reaction of K_2CO_3 (414.7 mg, 3.0 mmol), $Pd(OAc)_2$ (11.3 mg, 0.05 mmol), TFP (23.4 mg, 0.10 mmol), **2a** (267.8 mg, 1.2 mmol), **1i** (298.5 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.2 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3i** (yield: 265.6 mg, 60%) as a 1iquid {eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/

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16154169, 2016, 17, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201600406 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1) afforded impure **3i**, which was further purified with chromatography [eluent: dichloromethane (200 mL) to dichloromethane/ethyl acetate (60/1)]}. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.83–7.76 (m, 2H, ArH), 7.31–7.13 (m, 5H, ArH), 7.06 (d, J = 7.2 Hz, 1H, ArH), 6.84 (s, 1H, ArH), 5.24 (s, 2H, CH_2), 4.73 (s, 2H, CH_2), 3.70 (t, J = 5.6 Hz, 2H, CH_2), 2.83 (t, J = 5.3 Hz, 2H, CH_2), 2.40 (s, 3H, Me), 2.13 (s, 3H, Me), 1.87 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 142.8, 140.1, 138.8, 138.0, 137.7, 135.13, 135.07, 133.7, 133.2, 130.1, 129.5, 129.4, 128.6, 127.6, 126.8, 125.6, 74.2, 72.2, 62.1, 42.1, 30.8, 27.1, 21.3, 19.8; IR (neat): ν = 3060, 3015, 2985, 2944, 2925, 2869, 1760, 1598, 1494, 1469, 1366, 1320, 1268, 1248, 1151, 1107, 1091, 1063 cm^{-1}; MS (EI): m/z (%) = 447 (M^+, 29.40), 446 (100); HR-MS: m/z = 447.1869, calcd. for $C_{27}H_{29}NO_3S$ (M^+): 447.1868.

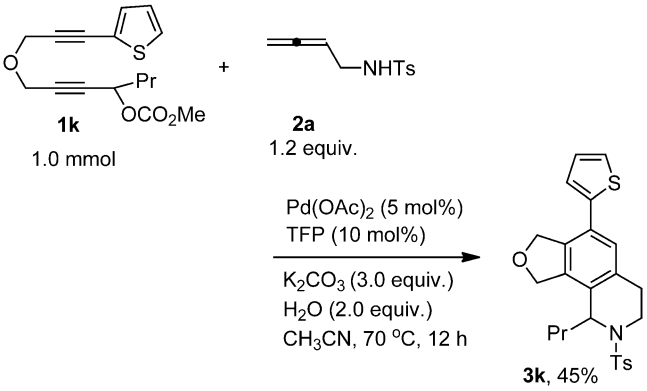
Preparation of 9,9-Dimethyl-4-(*m*-tolyl)-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3j)



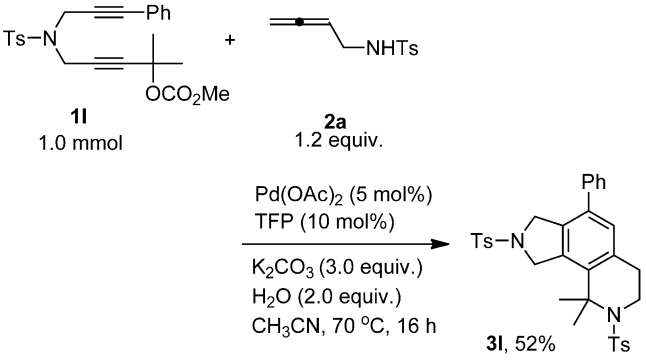
The reaction of K_2CO_3 (414.9 mg, 3.0 mmol), $Pd(OAc)_2$ (11.2 mg, 0.05 mmol), TFP (23.2 mg, 0.10 mmol), **2a** (268.5 mg, 1.2 mmol), **1j** (313.1 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (35.7 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3j** (yield: 295.1 mg, purity = 94%, 60%) as a solid {eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1) afforded impure **3j**, which was further purified with chromatography [eluent: dichloromethane (200 mL) to dichloromethane/ethyl acetate (60/1)]}; mp 145–146 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$):δ = 7.81 (d, J = 8.1 Hz, 2H, ArH), 7.34–7.33 (m, 3H, ArH), 7.19–7.08 (m, 3H, ArH), 7.00 (s, 1H, ArH), 5.26 (d, J = 12.3 Hz, 1H, one proton of CH_2), 5.19–5.00 (m, 3H, one proton of CH_2 and CH_2), 3.57 (t, J = 5.1 Hz, 2H, CH_2), 2.98–2.81 (m, 1H, one proton of CH_2), 2.78 (t, J = 5.1 Hz, 2H, and CH_2), 2.40 (s, 3H, Me), 2.37 (s, 3H, Me), 1.94–1.76 (m, 4H, one proton of CH_2 and Me), 0.73 (t, J = 7.2 Hz, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 142.8, 139.2, 138.1, 137.2, 135.9, 135.4, 135.2, 133.8, 129.4, 128.7, 128.4, 128.3, 128.1, 127.0, 124.7, 74.1 72.3, 66.5, 43.5, 32.4, 30.6, 27.0, 21.3, 9.1; IR (KBr): ν = 2980, 2933, 2908, 2869, 2837, 1596, 1466, 1324, 1303, 1159, 1128, 1091, 1064 cm^{-1}; MS (EI): m/z (%) = 461 (M^+, 0.03), 432 (100); anal. calcd. for $C_{28}H_{31}NO_3S$ (%): C 72.85, H 6.77, N 3.03; found: C 72.57, H 6.74, N 2.80.

Preparation of 9-Propyl-4-(thiophen-2-yl)-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3k)

The reaction of K_2CO_3 (413.8 mg, 3.0 mmol), $Pd(OAc)_2$ (11.3 mg, 0.05 mmol), TFP (23.3 mg, 0.10 mmol), **2a** (267.3 mg, 1.2 mmol), **1k** (319.9 mg, purity = 96%, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.2 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3k** (yield: 216.4 mg, purity = 95%, 45%) as a solid {eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1) afforded one part of pure **3k** and a mixture of **3k** and impurity, which was further purified by using chromatography [eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1)]}; mp 120–121 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.57 (d, J = 8.4 Hz, 2H, ArH), 7.28 (dd, J_1 = 5.1 Hz, J_2 = 1.2 Hz, 1H, ArH), 7.13–6.99 (m, 5H, ArH), 5.30–5.17 (m, 2H, CH_2), 5.13 (s, 2H, CH_2), 4.80 (dd, J_1 = 9.6 Hz, J_2 = 2.1 Hz, 1H, CH), 3.92–3.80 (m, 1H, one proton of CH_2), 3.60–3.46 (m, 1H, one proton of CH_2), 2.64–2.54 (m, 2H, CH_2), 2.27 (s, 3H, Me), 1.82–1.65 (m, 1H, one proton of CH_2), 1.64–1.38 (m, 3H, one proton of CH_2 + CH_2), 0.97 (t, J = 6.9 Hz, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 143.1, 141.7, 137.2, 136.4, 134.0, 132.4, 130.1, 129.2, 127.8, 126.85, 126.76, 126.7, 125.1, 124.5, 73.9, 72.2, 54.2, 38.2, 36.8, 25.4, 21.2, 19.3, 13.4; IR (KBr): ν = 2957, 2929, 2870, 1608, 1598, 1466, 1432, 1379, 1340, 1163, 1114, 1092, 1062 cm^{-1}; MS (EI): m/z (%) = 453 (M^+, 0.54), 410 (100); anal. calcd. for $C_{25}H_{27}NO_3S_2$ (%): C 66.19, H 6.00, N 3.09; found: C 66.10, H 6.05, N 2.90.



Preparation of 9,9-Dimethyl-4-phenyl-2,8-ditosyl-2,3,6,7,8,9-hexahydro-1*H*-pyrrolo[3,4-*h*]isoquinoline (3l)



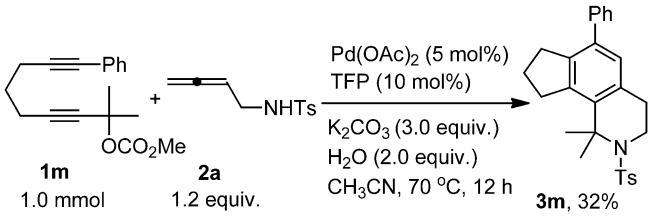
The reaction of K_2CO_3 (414.8 mg, 3.0 mmol), $Pd(OAc)_2$ (11.2 mg, 0.05 mmol), TFP (23.3 mg, 0.10 mmol), **2a** (267.2 mg, 1.2 mmol), **1l** (439.1 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.1 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 16 h afforded **3l** (yield: 303.4 mg, 52%) {eluent: petroleum ether/ethyl acetate/dichloromethane (5/1/1) afford-

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ed impure **3l**, which was further purified with chromatography [eluent: dichloromethane to dichloromethane/ethyl acetate (30/1)]}; mp 174–175 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78–7.70 (m, 4H, ArH), 7.45–7.21 (m, 9H, ArH), 6.96 (s, 1H, ArH), 4.73 (s, 2H, CH_2), 4.49 (s, 2H, CH_2), 3.66 (t, J = 5.4 Hz, 2H, CH_2), 2.82 (t, J = 5.0 Hz, 2H, CH_2), 2.37 (s, 6H, Me × 2), 1.84 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 143.8, 142.8, 140.0, 138.7, 138.5, 135.2, 134.3, 134.2, 132.5, 132.3, 129.7, 129.3, 128.5, 127.7, 127.6, 127.5, 126.6, 61.8, 54.4, 52.1, 41.7, 30.8, 27.0, 21.3, 21.2; IR (KBr): ν = 3060, 3027, 2988, 2923, 2867, 2841, 1597, 1494, 1468, 1400, 1348, 1333, 1316, 1266, 1248, 1163, 1092, 1067, 1017 cm^{-1}; MS (EI): *m/z* (%) 586 (M^+, 0.78), 571 (100); anal. calcd. for $C_{33}H_{34}N_2O_4S_2$ (%): C 67.55, H 5.84, N 4.77; found: C 67.71, H 5.90, N 4.63.

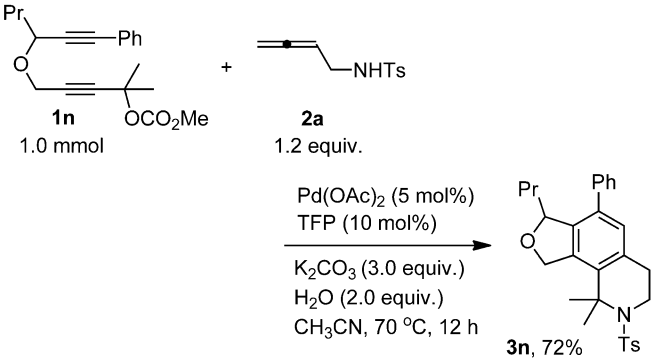
Preparation of 1,1-Dimethyl-6-phenyl-2-tosyl-2,3,4,7,8,9-hexahydro-1*H*-cyclopenta[*h*]isoquinoline (3m)



The reaction of K_2CO_3 (414.6 mg, 3.0 mmol), $Pd(OAc)_2$ (11.4 mg, 0.05 mmol), TFP (23.2 mg, 0.10 mmol), **2a** (267.2 mg, 1.2 mmol), **1m** (282.8 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.1 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3m** (yield: 135.3 mg, 32%) (eluent: petroleum ether/ethyl acetate = 10/1); mp 141–143 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.83–7.76 (m, 2H, ArH), 7.44–7.23 (m, 7H, ArH), 6.91 (s, 1H, ArH), 3.70–6.63 (m, 2H, CH_2), 3.08 (t, J = 7.1 Hz, 2H, CH_2), 2.88–2.78 (m, 4H, CH_2 × 2), 2.40 (s, 3H, Me), 2.04–1.91 (m, 8H, CH_2 + Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 142.7, 142.3, 140.9, 140.7, 140.5, 139.7, 136.4, 132.3, 129.4, 128.5, 128.2, 126.92, 126.88, 63.0, 42.0, 35.9, 32.2, 31.3, 27.0, 26.7, 21.4; IR (KBr): ν = 3057, 2947, 2869, 2840, 1597, 1495, 1463, 1443, 1396, 1366, 1319, 1287, 1265, 1248, 1157, 1111, 1092 cm^{-1}; MS (EI): *m/z* (%) = 431 (M^+, 0.48), 416 (100); anal. calcd. for $C_{27}H_{29}NO_2S$ (%): C 75.14, H 6.77, N 3.25; found: C 75.51, H 6.71, N 3.01.

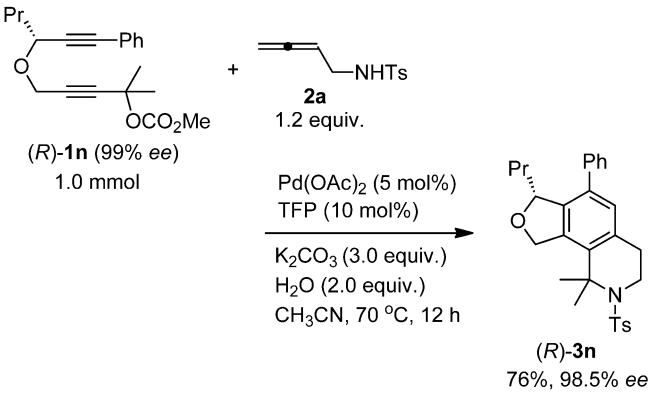
Preparation of 9,9-Dimethyl-4-phenyl-3-propyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3n)

The reaction of K_2CO_3 (414.2 mg, 3.0 mmol), $Pd(OAc)_2$ (11.0 mg, 0.05 mmol), TFP (23.0 mg, 0.10 mmol), **2a** (267.0 mg, 1.2 mmol), **1n** (329.2 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.2 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3n** (yield: 343.5 mg, 72%) as a solid after chromatography on silica gel]eluent: petroleum ether/ethyl acetate/dichloromethane (7/1/1)]; mp 130–131 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, J = 8.4 Hz, 2H, ArH), 7.46–7.20 (m, 7H, ArH), 6.94 (s, 1H, ArH), 5.55–5.44 (m, 1H, CH), 5.28–5.12 (m, 2H, CH_2),



3.78–3.62 (m, 2H, CH_2), 2.93–2.76 (m, 2H, CH_2), 2.40 (s, 3H, Me), 1.862 (s, 3H, Me), 1.858 (s, 3H, Me), 1.30–1.05 (m, 4H, CH_2 × 2), 0.66 (t, J = 6.9 Hz, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 142.7, 140.2, 139.9, 139.5, 137.7, 135.8, 134.6, 133.6, 130.0, 129.4, 128.4, 128.0, 127.3, 126.7, 82.3, 72.6, 62.1, 42.0, 35.5, 30.8, 27.0, 26.9, 21.3, 18.1, 13.5; IR (KBr): ν = 3057, 2956, 2929, 2870, 1598, 1495, 1465, 1400, 1367, 1322, 1268, 1248, 1150, 1130, 1092 cm^{-1}; MS (EI): *m/z* (%) = 475 (M^+, 2.02), 432 (100); anal. calcd. for $C_{29}H_{33}NO_3S$ (%): C 73.23, H 6.99, N 2.94; found: C 73.38, H 7.04, N 2.75.

Preparation of (*R*)-9,9-Dimethyl-4-phenyl-3-propyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline [(*R*)-3n]



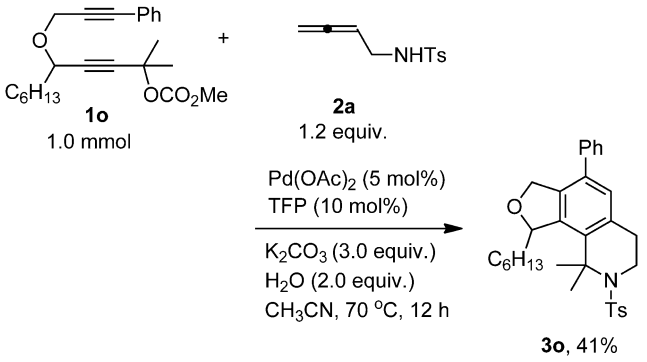
The reaction of K_2CO_3 (413.8 mg, 3.0 mmol), $Pd(OAc)_2$ (11.2 mg, 0.05 mmol), TFP (23.4 mg, 0.10 mmol), **2a** (267.6 mg, 1.2 mmol), (*R*)-**1n** (327.9 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.0 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded (*R*)-**3n** (yield: 363.2 mg, 76%) as a solid after chromatography on silica gel {eluent: petroleum ether/ethyl acetate/dichloromethane (7/1/1) afforded a part of pure (*R*)-**3n** and a mixture of (*R*)-**3n** and impurity, which was further purified by using the eluent [petroleum ether/ethyl acetate/dichloromethane (7/1/1)]}; 98.5% *ee*; HPLC (conditions: Chiralcel IA-H column, hexane/*i*-PrOH = 80/20, 1.0 mL min^{-1}, λ = 214 nm): t_R(minor) = 11.2 min, t_R(major) = 11.9 min); $[\alpha]_D^{20}$: −13.8 (c = 1.095, $CHCl_3$); mp 131–133 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.78 (d, J = 8.4 Hz, 2H, ArH), 7.45–7.22 (m, 7H, ArH), 6.93 (s, 1H,

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ArH), 5.55–5.44 (m, 1H, CH), 5.28–5.12 (m, 2H, CH_2), 3.79–3.62 (m, 2H, CH_2), 2.93–2.76 (m, 2H, CH_2), 2.41 (s, 3H, Me), 1.860 (s, 3H, Me), 1.856 (s, 3H, Me), 1.30–1.03 (m, 4H, $CH_2\times2$), 0.65 (t, $J=7.2$ Hz, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta=142.8$, 140.2, 139.9, 139.5, 137.8, 135.8, 134.7, 133.6, 130.0, 129.4, 128.4, 128.0, 127.3, 126.7, 82.3, 72.6, 62.1, 42.0, 35.5, 30.8, 27.02, 26.95, 21.3, 18.1, 13.6; IR (KBr): $\nu=3057$, 2956, 2929, 2870, 1598, 1465, 1367, 1322, 1268, 1248, 1150, 1130, 1092 cm^{-1}; MS (EI): m/z (%) = 475 (M^+, 4.97), 432 (100); anal. calcd. for $C_{29}H_{33}NO_3S$ (%): C 73.23, H 6.99, N 2.94; found: C 73.24, H 7.04, N 2.73.

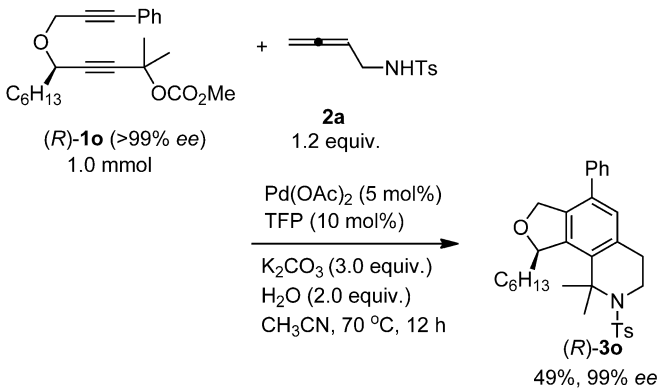
Preparation of 1-Hexyl-9,9-dimethyl-4-phenyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3o)



The reaction of K_2CO_3 (413.9 mg, 3.0 mmol), $Pd(OAc)_2$ (11.3 mg, 0.05 mmol), TFP (23.4 mg, 0.10 mmol), **2a** (269.0 mg, 1.2 mmol), **1o** (370.9 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.2 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3o** (yield: 213.4 mg, 41%) as a liquid after chromatography on silica gel [eluent: petroleum ether/ethyl acetate/dichloromethane (10/1/1)]: ^{1}H NMR (300 MHz, $CDCl_3$): $\delta=7.74$ (d, $J=8.1$ Hz, 2H, ArH), 7.50–7.19 (m, 7H, ArH), 6.97 (s, 1H, ArH), 5.44–5.32 (m, 1H, CH), 5.16 (d, $J=12.9$ Hz, 1H, one proton of CH_2), 4.76 (d, $J=13.2$ Hz, 1H, one proton of CH_2), 4.02–3.87 (m, 1H, one proton of CH_2), 3.64–3.49 (m, 1H, one proton of CH_2), 3.01–2.62 (m, 2H, CH_2), 2.39 (s, 3H, Me), 1.92 (s, 3H, Me), 1.90 (s, 3H, Me), 1.78–1.58 (m, 3H, one proton of CH_2 and CH_2), 1.45–1.20 (m, 7H, one proton of CH_2 and $CH_2\times3$), 1.00–0.78 (m, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta=$ 142.7, 141.1, 140.0, 139.3, 137.2, 137.1, 134.4, 134.1, 129.3, 128.9, 128.5, 127.7, 127.4, 126.8, 84.5, 69.0, 62.7, 42.2, 34.6, 31.6, 30.6, 29.8, 29.5, 28.9, 27.0, 22.5, 21.3, 14.0; IR (neat): $\nu=2953$, 2927, 2855, 1598, 1495, 1462, 1365, 1325, 1266, 1248, 1160, 1130, 1091, 1056 cm^{-1}; MS (EI): m/z (%) = 517 (M^+, 1.14), 432 (100); HR-MS: $m/z=517.2647$, calcd. for $C_{32}H_{39}NO_3S$ (M^+): 517.2651.

Preparation of (*R*)-1-Hexyl-9,9-dimethyl-4-phenyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline [(*R*)-3o]

The reaction of K_2CO_3 (414.3 mg, 3.0 mmol), $Pd(OAc)_2$ (11.3 mg, 0.05 mmol), TFP (23.2 mg, 0.10 mmol), **2a** (268.1 mg, 1.2 mmol), (*R*)-**1o** (368.6 mg, 1.0 mmol)/CH_3CN (8.0 mL), and H_2O (36.1 mg, 2.0 mmol)/CH_3CN (2.0 mL) at



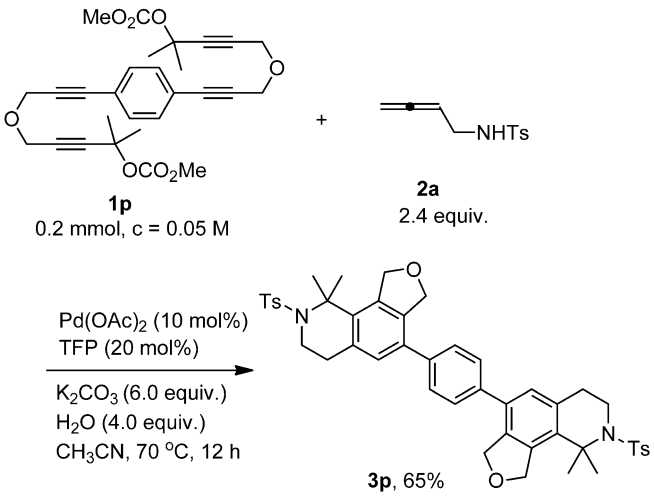
70 °C for 12 h afforded (*R*)-**3o** (yield: 253.7 mg, 49%) as a liquid after chromatography on silica gel {eluent: eluent: petroleum ether/ethyl acetate/dichloromethane (10/1/1) afforded a part of pure (*R*)-**3o** and a mixture of (*R*)-**3o** and impurity, which was further purified by using the eluent [petroleum ether/ethyl acetate/dichloromethane (12/1/1)]}; 99% *ee*; HPLC (conditions: Chiralcel AD-H column, hexane/*i*-PrOH = 80/20, 1.0 mL min^{-1}, $\lambda=214$ nm): t_R(major) = 10.0 min, t_R(minor) = 13.3 min; $[\alpha]_D^{20}$: +78.3 ($c=0.94$, $CHCl_3$). ^{1}H NMR (300 MHz, $CDCl_3$): $\delta=7.74$ (d, $J=8.4$ Hz, 2H, ArH), 7.47–7.30 (m, 5H, ArH), 7.28–7.21 (m, 2H, ArH), 6.96 (s, 1H, ArH), 5.43–5.33 (m, 1H, CH), 5.16 (d, $J=12.6$ Hz, 1H, one proton of CH_2), 4.76 (d, $J=12.9$ Hz, 1H, one proton of CH_2), 3.94 (dt, $J_1=13.5$ Hz, $J_2=4.8$ Hz, 1H, one proton of CH_2), 3.64–3.50 (m, 1H, one proton of CH_2), 3.00–2.87 (m, 1H, one proton of CH_2), 2.80 (dt, $J_1=$ 16.2 Hz, $J_2=4.1$ Hz, 1H, one proton of CH_2), 2.39 (s, 3H, Me), 1.92 (s, 3H, Me), 1.90 (s, 3H, Me), 1.78–1.20 (m, 10H, one proton of CH_2 and $CH_2\times3$), 0.89 (t, $J=6.8$ Hz, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta=142.7$, 141.1, 140.0, 139.3, 137.2, 137.1, 134.4, 134.1, 129.3, 129.0, 128.5, 127.8, 127.4, 126.8, 84.5, 69.0, 62.7, 42.2, 34.6, 31.6, 30.7, 29.8, 29.5, 28.9, 27.0, 22.5, 21.3, 14.0; IR (neat): $\nu=2956$, 2926, 2855, 1596, 1462, 1366, 1323, 1160, 1130, 1091, 1056 cm^{-1}; MS (EI): m/z (%) = 517 (M^+, 3.6), 432 (100); HR-MS: $m/z=$ 517.2663, calcd. for $C_{32}H_{39}NO_3S$ (M^+): 517.2651.

Preparation of 1,4-Bis(9,9-dimethyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinolin-4-yl)benzene (3p)

To a flame-dried Schlenk tube containing K_2CO_3 (165.6, 1.20 mmol) were added $Pd(OAc)_2$ (4.6 mg, 0.02 mmol), TFP (9.5 mg, 0.04 mmol), **2a** (109.1 mg, 0.48 mmol), **1p** (98.8 mg, 0.2 mmol)/CH_3CN (3.0 mL), and H_2O (14.3 mg, 0.8 mmol)/CH_3CN (1.0 mL) sequentially under nitrogen atmosphere. The reaction was complete after being stirred at 70 °C for 12 h as monitored by TLC (eluent: dichloromethane/ethyl acetate = 20/1). After cooling to the room temperature, the resulting mixture was diluted with dichloromethane (30 mL), filtered through a short silica gel, and eluted with dichloromethane (20 mL). After evaporation of the solvent, the yields of product **3p** and **4p** were determined by ^{1}H NMR analysis of the crude product with mesitylene (18.4 μL) as the internal standard. Chromatography on silica gel (eluent: dichloromethane/ethyl acetate = 400/20) afforded **3p** as a solid; yield: 102.1 mg (65%); mp 270 °C ($CHCl_3$). ^{1}H NMR (300 MHz, $CDCl_3$): $\delta=7.78$ (d, $J=8.4$ Hz, 4H,

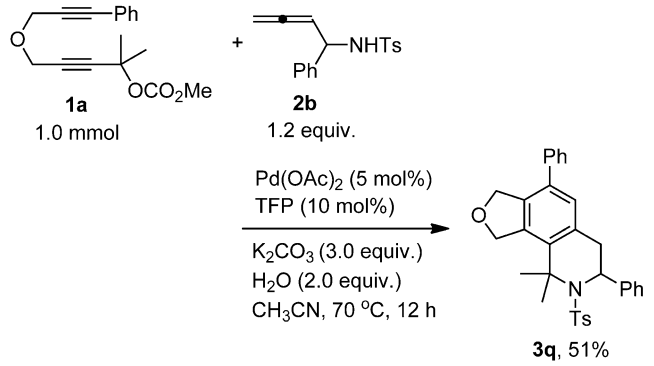
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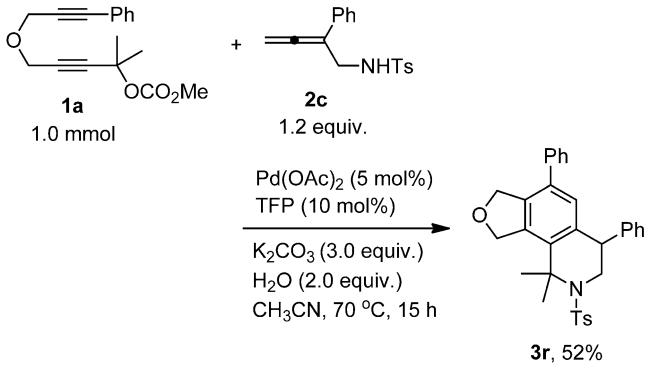
ArH), 7.39 (s, 4H, ArH), 7.34–7.22 (m, 4H, ArH), 7.07 (s, 2H, ArH), 5.24 (s, 4H, $CH_2\times2$), 5.10 (s, 4H, $CH_2\times2$), 3.71 (t, $J=5.4$ Hz, 4H, $CH_2\times2$), 2.89 (t, $J=5.1$ Hz, 4H, $CH_2\times2$), 2.41 (s, 6H, Me×2), 1.86 (s, 12H, Me×4); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta=143.0$, 140.3, 138.7, 138.4, 137.5, 136.3, 134.1, 133.4, 129.6, 129.1, 128.1, 126.9, 74.1, 72.4, 62.2, 42.2, 31.1, 27.2, 21.5; IR (KBr): $\nu=3003$, 2944, 2920, 2858, 2837, 1465, 1385, 1320, 1149, 1132, 1091, 1063 cm^{-1}; MS (MALDI): $m/z=789$ ($M+H^+$); HR-MS: $m/z=789.3034$, calcd. for $C_{46}H_{49}N_2O_6S_2$ ($M+H^+$): 789.3027.

Preparation of 9,9-Dimethyl-4,7-diphenyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3q)



The reaction of K_2CO_3 (413.8 mg, 3.0 mmol), $Pd(OAc)_2$ (11.3 mg, 0.05 mmol), TFP (23.1 mg, 0.10 mmol), **1a** (285.3 mg, 1.0 mmol)/CH_3CN (5.0 mL), **2b** (358.5 mg, 1.2 mmol)/CH_3CN (3.0 mL), and H_2O (36.2 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 12 h afforded **3q** (yield: 259.6 mg, 51%) as a solid {eluent: eluent: petroleum ether/ethyl acetate/dichloromethane (7/1/1) for the first round to get a part of pure **3q** and a mixture of **3q** and impurity, which was further purified by using the elutent [petroleum ether/ethyl acetate/dichloromethane (7/1/1)]}; mp 197–199 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): $\delta=$ 7.78 (d, $J=8.1$ Hz, 2H, ArH), 7.45–7.04 (m, 13H, ArH), 5.85–5.77 (m, 1H, CH), 5.16–4.88 (m, 4H, $CH_2\times2$), 3.41 (dd, $J_1=15.9$ Hz, $J_2=2.7$ Hz, 1H, one proton of CH_2), 3.31 (dd, $J_1=15.8$ Hz, $J_2=5.0$ Hz, 1H, one proton of CH_2), 2.36 (s, 3H, Me), 1.76 (s, 3H, Me), 1.70 (s, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta=142.6$, 140.9, 139.9, 139.3, 137.8, 137.5, 135.4, 134.1, 131.7, 129.4, 129.0, 128.5, 128.0, 127.9, 127.7, 127.5, 126.8, 126.7, 74.0, 72.4, 62.2, 55.4, 33.5, 29.8, 29.2, 21.3; IR (KBr): $\nu=3053$, 3030, 3004, 2959, 2911, 2855, 1596, 1495, 1465, 1448, 1381, 1323, 1249, 1164, 1126, 1087, 1074 cm^{-1}; MS (EI): m/z (%) = 494 ($(M-CH_3)^+$, 100); anal. calcd. for $C_{32}H_{31}NO_3S$ (%): C 75.41, H 6.13, N 2.75; Found: C 75.11, H 6.22, N 2.65.

Preparation of 9,9-Dimethyl-4,6-diphenyl-8-tosyl-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3r)



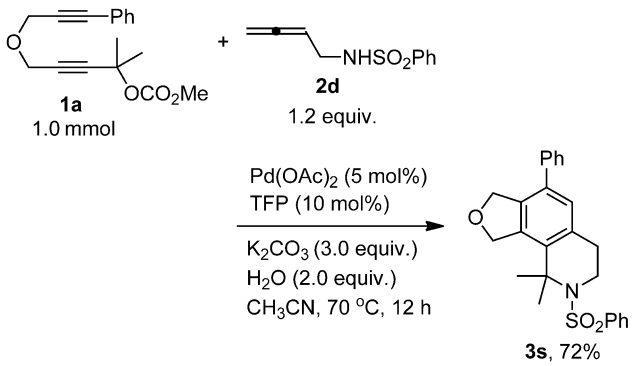
The reaction of K_2CO_3 (415.1 mg, 3.0 mmol), $Pd(OAc)_2$ (11.4 mg, 0.05 mmol), TFP (23.1 mg, 0.10 mmol), **1a** (286.6 mg, 1.0 mmol)/CH_3CN (6.0 mL), **2c** (359.4 mg, 1.2 mmol)/CH_3CN (2.0 mL), and H_2O (36.2 mg, 2.0 mmol)/CH_3CN (2.0 mL) at 70 °C for 15 h afforded **3r** (yield: 265.2 mg, 52%) as a solid {eluent: petroleum ether/ethyl acetate/dichloromethane (7/1/1) afforded impure **3r**, which was further purified with chromatography [eluent: dichloromethane (150 mL) to dichloromethane/ethyl acetate (50/1)]}; mp 210–212 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): $\delta=$ 7.38–7.20 (m, 7H, ArH), 7.19–7.06 (m, 3H, ArH), 6.95 (d, $J=8.1$ Hz, 2H, ArH), 6.92–6.84 (m, 3H, ArH), 5.38–5.25 (m, 2H, CH_2), 5.16–5.02 (m, 2H, CH_2), 5.20 (t, $J=3.8$ Hz, 1H, CH), 3.86 (dd, $J_1=13.2$ Hz, $J_2=4.2$ Hz, 1H, one proton of CH_2), 3.80 (dd, $J_1=13.5$ Hz, $J_2=3.9$ Hz, 1H, one proton of CH_2), 2.32 (s, 3H, Me), 1.96 (s, 3H, Me), 1.95 (s, 3H, Me); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta=142.6$, 142.3 139.2, 138.9, 138.5, 138.0, 136.0, 135.4, 134.2, 130.1, 129.0, 128.55, 128.47, 128.2, 127.7, 127.5, 127.0, 126.3, 74.1, 72.4, 62.3, 48.1, 45.7, 27.4, 25.7, 21.3; IR (KBr): $\nu=3059$, 3026, 2946, 2917, 2863, 1599, 1494, 1467, 1452, 1368, 1321, 1303, 1266, 1240, 1155, 1129, 1091, 1072, 1030 cm^{-1}; MS (EI): m/z (%) = 509 (M^+, 0.13), 494 [$(M-CH_3)^+$, 100]; anal. calcd. for $C_{32}H_{31}NO_3S$ (%): C 75.41, H 6.13, N 2.75; found: C 75.07, H 6.16, N 2.54.

Preparation of 9,9-Dimethyl-4-phenyl-8-(phenylsulfonyl)-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3s)

The reaction of K_2CO_3 (414.6 mg, 3.0 mmol), $Pd(OAc)_2$ (11.2 mg, 0.05 mmol), TFP (23.3 mg, 0.10 mmol), **1a** (285.6 mg, 1.0 mmol)/CH_3CN (5.0 mL), **2d** (251.2 mg,

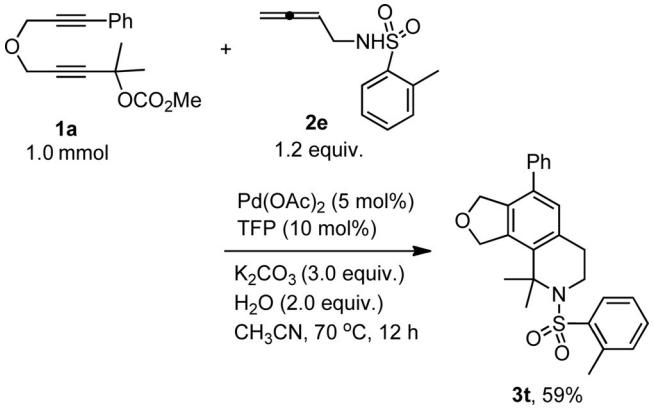
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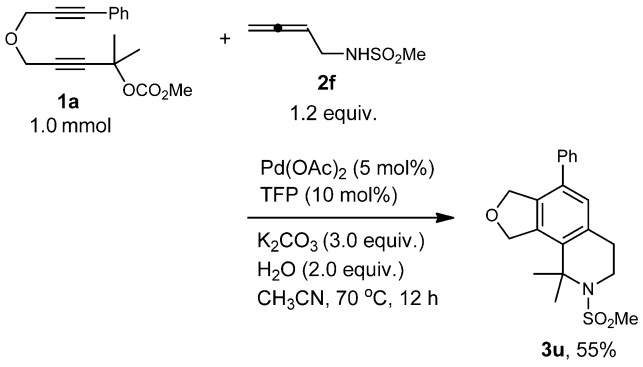
1.2 mmol)/CH_3CN (4.0 mL), and H_2O (36.2 mg, 2.0 mmol)/CH_3CN (1.0 mL) at 70 °C for 12 h afforded **3s** (yield: 302.4 mg, 72%) as a solid {eluent: petroleum ether/ethyl acetate/dichloromethane (5/1/1) afforded impure **3s**, which was further purified with chromatography [eluent: dichloromethane (100 mL) to dichloromethane/ethyl acetate (60/1)]}; mp 167–169 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.93–7.86 (m, 2H, ArH), 7.57–7.27 (m, 8H, ArH), 7.04 (s, 1H, ArH), 5.22 (s, 2H, CH_2), 5.05 (s, 2H, CH_2), 3.73 (t, J = 5.6 Hz, 2H, CH_2), 2.88 (t, J = 5.4 Hz, 2H, CH_2), 1.85 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 143.1, 139.3, 137.8, 137.4, 136.0, 134.0, 133.8, 132.1, 129.1, 128.9, 128.6, 127.7, 127.5, 126.7, 74.0, 72.3, 62.2, 42.1, 30.9, 27.1; IR (KBr): ν = 3072, 3051, 2997, 2939, 2908, 2894, 2866, 2836, 1468, 1446, 1367, 1329, 1308, 1251, 1165, 1148, 1129, 1090, 1061 cm^{-1}; MS (EI): *m/z* (%) = 419 (M^+, 2.4), 404 (100); anal. calcd. for $C_{25}H_{25}NO_3S$ (%): C 71.57, H 6.01, N 3.34; found: C 71.37, H 6.04, N 3.32.

Preparation of 9,9-Dimethyl-4-phenyl-8-(*o*-tolylsulfonyl)-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3t)



The reaction of K_2CO_3 (414.3 mg, 3.0 mmol), $Pd(OAc)_2$ (11.2 mg, 0.05 mmol), TFP (23.4 mg, 0.10 mmol), **1a** (285.7 mg, 1.0 mmol)/CH_3CN (5.0 mL), **2e** (268.0 mg, 1.2 mmol)/CH_3CN (4.0 mL), and H_2O (36.5 mg, 2.0 mmol)/CH_3CN (1.0 mL) at 70 °C for 12 h afforded **3t** (yield: 256.5 mg, 59%) as a solid {eluent: petroleum ether/ethyl acetate/dichloromethane (6/1/1) afforded impure **3t**, which was further purified with chromatography [eluent: dichloromethane (100 mL) to dichloromethane/ethyl acetate (50/1)]}; mp 138–139 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.05–7.97 (m, 1H, ArH), 7.45–7.22 (m, 8H, ArH), 7.06 (s, 1H, ArH), 5.21 (s, 2H, CH_2), 5.05 (s, 2H, CH_2), 3.72 (t, J = 5.9 Hz, 2H, CH_2), 2.95 (t, J = 5.6 Hz, 2H, CH_2), 2.68 (s, 3H, Me), 1.82 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 140.5, 139.2, 137.6, 137.2, 137.1, 135.7, 134.0, 133.4, 132.5, 132.4, 129.2, 129.1, 128.4, 127.6, 127.4, 125.9, 73.9, 72.2, 62.2, 41.3, 30.3, 26.9, 20.0; IR (KBr): ν = 3057, 3021, 2994, 2970, 2935, 2861, 2840, 1469, 1314, 1251, 1157, 1128, 1065cm^{-1}; MS (EI): *m/z* (%) = 433 (M^+, 2.4), 418 (100); anal. calcd. for $C_{26}H_{27}NO_3S$ (%): C 72.03, H 6.28, N 3.23; found: C 71.84, H 6.24, N 3.10.

Preparation of 9,9-Dimethyl-4-phenyl-8-(methylsulfonyl)-1,3,6,7,8,9-hexahydrofuro[3,4-*h*]isoquinoline (3u)



The reaction of K_2CO_3 (414.1 mg, 3.0 mmol), $Pd(OAc)_2$ (11.4 mg, 0.05 mmol), TFP (23.4 mg, 0.10 mmol), **1a** (285.2 mg, 1.0 mmol)/CH_3CN (5.0 mL), **2f** (177.0 mg, 1.2 mmol)/CH_3CN (4.0 mL), and H_2O (36.1 mg, 2.0 mmol)/CH_3CN (1.0 mL) at 70 °C for 12 h afforded **3u** (yield: 196.0 mg, 55%) as a solid [eluent: petroleum ether/ethyl acetate/dichloromethane (3/1/1)]; mp 188–189 °C (*n*-hexane/DCM). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.47–7.29 (m, 5H, ArH), 7.08 (s, 1H, ArH), 5.26 (s, 2H, CH_2), 5.07 (s, 2H, CH_2), 3.68 (t, J = 5.4 Hz, 2H, CH_2), 3.01 (s, 3H, Me), 2.96 (t, J = 5.4 Hz, 2H, CH_2), 1.88 (s, 6H, Me × 2); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 139.2, 137.7, 137.5, 136.0, 134.0, 133.9, 129.0, 128.5, 127.6, 127.5, 73.9, 72.3, 61.8, 43.2, 42.3, 31.2, 27.1; IR (KBr): ν = 3051, 3012, 2950, 2933, 2861, 2840, 1468, 1367, 1316, 1269, 1251, 1160, 1137, 1128, 1068 cm^{-1}; MS (EI): *m/z* (%) = 357 (M^+, 8.2), 342 (100); anal. calcd. for $C_{20}H_{23}NO_3S$ (%): C 67.20, H 6.49, N 3.92; found: C 66.95, H 6.44, N 3.77.

Acknowledgements

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[12] Crystal data for compound **3e**: $C_{30}H_{27}NO_3S$, MW = 481.59, monoclinic, space group *P 21/n*, final *R* indices [I > 2σ(I)], *R1* = 0.0486, *wR2* = 0.1097, *R* indices (all data) *R1* = 0.0791, *wR2* = 0.0.1261, *a* = 15.7071(15) Å, *b* = 9.4810(7) Å, *c* = 17.8828(18) Å, *α* = 90.00°, *β* = 111.590(12)°, *γ* = 90.00°, *V* = 2476.2(4) Å^3, *T* = 293(2) K, *Z* = 4, reflections collected/unique 10217/4527 (R_{int} = 0.0322), number of observations [> 2σ(I)] 3049, parameters: 317. CCDC 1444531 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre *via* www.ccdc.cam.ac.uk/data_request/cif.

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